Consequence-Aware Failure Diagnostics for Learned Particle-Reconstruction Representations

Collaboration concept note

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Abstract

A reconstruction model can keep good average accuracy while its internal representation responds badly to a change in measurement conditions. We propose a controlled, replay-based study of frozen learned representations under declared interventions, asking whether they can separate corrupted acquisition from clean-but-under-supported physics, and whether an alarm means anything scientifically. The central test is whether alarms identify the declared failure family and whether alarm magnitude is an adequate proxy for downstream scientific loss. We explicitly quantify two discordant regimes: detectable but scientifically harmless change, and consequential degradation without a strong alarm. Two testbeds are used—public neutrino-telescope point clouds for realism, and a waveform task whose noise covariance we generate, so that assumed and realized covariance are both known and consequence can be computed rather than assumed. Layerwise representation evidence is compared against input tests, MMD, classifier two-sample tests, embedding distance, uncertainty, and output-only monitoring at a matched false-alert budget, with abstention on families the study never declared.

1 Gap and contribution

Existing shift monitors can detect changes in inputs, outputs, or embeddings, but a scientific reconstruction pipeline requires two additional distinctions: whether the change reflects corrupted acquisition or valid but under-supported physics, and whether the detected change materially degrades a physics result. We test whether intermediate representations improve this attribution beyond generic monitoring at the same false-alert budget, while allowing abstention on ambiguous or unseen failures.

In waveform reconstruction, structured acquisition noise induces a likelihood geometry through the inverse covariance: the correct residual weighting is Σ^{-1}, and a model evaluated under the wrong weighting can prefer reconstructions that are worse in the metric the instrument implies. Architecture and training coverage further determine which physically relevant distinctions survive into the representation. This study asks whether violations of these assumptions can be diagnosed after training, and whether a detected violation is scientifically consequential. The artifact is a reproducible experiment suite—configurations, adapters, hooks, replay scripts, and reports—not a maintained software product.

Identifiability limit. Diagonalizing a covariance yields decorrelated statistical modes, not labeled physical sources: $\Sigma = \Sigma_1 + \Sigma_2$ admits infinitely many positive-semidefinite splits, and low-rank factors are rotation-degenerate. All attribution here is over *declared intervention families* under a stated simulator. Observing a response profile does not identify the cause of an unknown real shift, and we make no such claim.

2 Failure contracts

Three declared labels, frozen before evaluation.

N—acquisition/noise shift. PSD or covariance change, jitter, drift, spectral lines, glitches, tails, hit thinning, channel or module loss. The measurement is corrupted.

S—training-support shift. Window or event composition moves toward physical strata that are sparse under the clean training distribution. The events themselves are clean and physically valid; only the support has moved.

E—evaluation-contract fault. Wrong units, coordinate ordering, output semantics, PSD convention, or metric implementation. This is a pipeline bug, not a distribution shift.

N versus S is the scientific core. E is decided by deterministic schema and range invariants, so it is reported as a gate result, not as a class in a classification score. The informative and difficult separation is N from S. N and S may both move the observed input distribution, so generic input-shift detection alone cannot distinguish them. Evidence for N comes from acquisition-quality indicators, paired corruption interventions, or known violations of the measurement model. S is defined by physically valid, uncorrupted events that occupy low-density regions of the clean training support. The scientific problem is therefore not whether the input changed, but whether the change reflects corrupted measurement or valid but under-supported physics. That distinction is what matters operationally: discarding S rejects physically valid events, whereas rejecting N removes measurements that violate the declared acquisition contract.

3 Claims

C1—Detection. At a 1% false-alert rate on nominal, non-overlapping windows, the best monitoring signature detects at least 80% of held-out abrupt and gradual N/S shifts within five windows, and is non-inferior to the strongest eligible baseline by a five-point power margin. Delay is primary in *windows*.

C2—Attribution with abstention. Primary endpoint is macro-F1 over {N, S} on held-out cases, with N+S mixtures scored by multi-label F1 in the same endpoint.

A split-conformal nonconformity score [4] is used to *abstain* on cases that do not resemble the declared training families. Unseen-family detection is evaluated by AUROC, while attribution quality under abstention is reported through selective

risk and macro-F1 as functions of retained coverage [5]. The selective endpoint is risk–coverage AUC, with N/S macro-F1 at 80% retained coverage reported alongside it. Detection of the unknown and attribution among the known are thus not scored as if they were the same quantity.

The paired difference ΔF1 between the full layerwise and all-generic arms is reported with a 95% confidence interval, and read three ways. A practically meaningful benefit requires the interval to exclude zero *and* the point estimate to reach 0.10 macro-F1; equivalence is supported if the interval lies within a predeclared ± 0.05 margin; otherwise the result is inconclusive. This lets a strong generic baseline produce a meaningful negative result — internal hooks may not justify their cost.

C3—Cost. Ten-percent event sampling plus fixed-size sketches yields p95 replay-path latency overhead $< 10\%$, $\geq 10\times$ less representation storage, and ≤ 5 points of power or F1 loss relative to full capture.

C4—Is alarm magnitude an adequate proxy for scientific consequence? This is posed as a test of adequacy, not as a prediction that the two agree or disagree. Let A be the monitor's alarm magnitude and K the downstream scientific loss for the same intervention cell. High consequence is $K \geq \kappa_m$, where κ_m is a testbed-specific scientific tolerance frozen before evaluation — provisionally a 10% degradation for the waveform amplitude task, and for angular or clustering metrics a physically meaningful change in resolution, efficiency, purity, or ARI rather than a generic percentage.

Table 1: C4 endpoints. Either outcome is informative.

Primary	Within-family Spearman $\rho(A, K)$; AUROC/AUPRC for detecting $K \geq \kappa_m$; low-alarm/high-loss and high-alarm/low-loss rates
Secondary	Amplitude bias, calibration error, subspace angle, neighbour retention
Agreement	Alarms reliably rank consequence and detect high-consequence cells; alarm magnitude is usable as a severity indicator
Discordance	Alarm magnitude fails to predict consequence, or the off-diagonal rates are materially large; consequence must then be monitored separately from detection
Inconclusive	Intervals too wide at the achieved sample size; reported as such, not resolved by post hoc thresholds

The waveform testbed decides this, because the generator supplies both the assumed $\hat{\Sigma}$ used by the encoder and the realized Σ, permitting a controlled test of whether alarms correspond to degradation in the correct likelihood-weighted residual. C4 is primary there and secondary on the point-cloud testbed, where consequence is downstream reconstruction error and no exact covariance exists. We state that asymmetry rather than implying symmetric coverage.

4 Testbeds and monitoring design

Neutrino-telescope point clouds (realism). One primary public configuration with a released checkpoint exposing a 128-dimensional pre-head event representation. Module dropout, hit thinning, and time jitter instantiate N; S windows use clean low-density strata in energy, topology, vertex/containment, and multiplicity, estimated from training data only. One secondary architecture serves as a consistency check.

Waveforms (mechanism). Non-overlapping windows at fixed sampling rate with a noise-aware linear encoder and amplitude reconstruction. The generator constructs PSD/CSD models, samples seeded frequency-domain coefficients with the required real-signal symmetry, and adds stationary color, piecewise nonstationarity, lines, drift, glitches, and tails to clean signals, validating units and second-order spectral statistics. It is a controlled intervention engine, not another dataset, reconstruction model, or physical-source separator.

Monitoring comparison. Each window records input-quality summaries, named-layer moments or sketches, outputs, uncertainty, delayed task metrics, and calibration. A regularized logistic classifier is fitted with identical splits to five arms: input only; output/uncertainty; final embedding only; all generic-monitor outputs; and the full layerwise signature. The fourth is the honest strong baseline, committed to in advance. Eligible baselines — corrected univariate input tests, RBF-MMD [6], a classifier two-sample test [7], embedding mean/covariance distance, output tests, uncertainty/entropy — receive identical windows and false-alert budgets. Clean miscalibration is a model result, not automatically a software bug [8]; latent displacement is normalized by clean within-class dispersion. Held-out severities, seeds, support strata, mixtures, and one unseen family prevent perturbation lookup; windows and thresholds are frozen after development; intervals resample event groups and perturbation seeds separately; null results require a passed positive control, adequate precision, and a predeclared equivalence margin.

Table 2: Core experiment matrix. C1–C3 run on both testbeds; C4 is primary on waveforms.

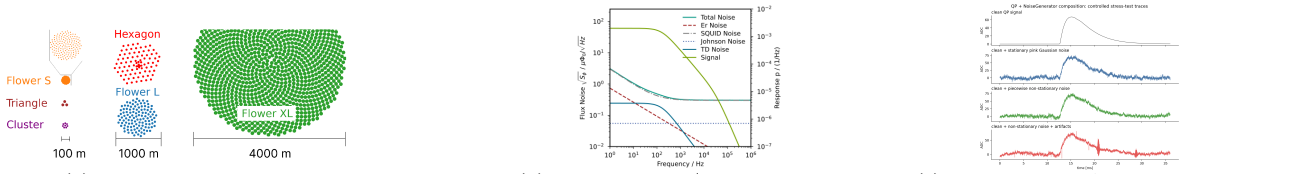
ID	Test	Primary endpoints
E0	Clean replay; deliberate E faults	Reference discrepancy; gate sensitivity/specificity
E1	Held-out abrupt/-gradual N/S	Power at 1% false alert; AUROC/AUPRC; delay in windows
E2	N/S attribution, mixtures, unseen family	Macro/multi-label F1 over {N,S}; paired CI vs generic arm; unknown AUROC
E3	Capture vs sample/sketch	p50/p95 latency; bytes/event; power/F1 loss
E4	Alarm vs consequence (C4)	$\rho(A, K)$; AUROC for $K \geq \kappa_m$; off-diagonal rates

5 Optional extensions

Extension G1 — detector geometry (conditional). If common latent events can be simulated and passed through two detector geometries under matched acquisition interventions, we will estimate whether geometry modifies robustness. This arm begins only after a pairing-and-matching gate is passed. Without paired simulation, cross-geometry results will be reported only as consistency comparisons and will not support causal layout claims. Matching variables, precision targets, and the estimator are deferred to a separate protocol note rather than fixed here.

Possible joint validation in hierarchical reconstruction

A natural external validation is a hierarchical particle-reconstruction pipeline such as SPINE/Graph-SPICE [2]. Unlike an event-level embedding, Graph-SPICE provides learned *point* representations whose geometry directly controls particle-instance clustering — a level of the reconstruc-



(a) detector geometries (b) detector signal/noise spectrum (c) generated waveform stress tests

Figure 1: Complementary testbeds. (a) Six top-view geometries, reproduced from NuBench Fig. 3 [1]. (b) The supplied Al_2O_3/Al athermal-detector response and component PSDs. (c) The generator composes a clean optimal-filter-like pulse with stationary colored noise, piecewise nonstationarity, and explicit artifacts. Panels (b)–(c) are project outputs.

tion chain that the point-cloud direction task above does not exercise at all. A compact joint arm could test whether one frozen diagnostic conclusion replicates under controlled hit dropout, coordinate jitter, module loss, or artificial noise, with particular attention to ambiguous regions near interaction vertices.

Relation to chained-model uncertainty propagation. Douglas et al. [3] study how input uncertainty propagates through a chained neutrino-LAr reconstruction, using synthetic noise to compare an uncertainty-aware model against an uncertainty-blinded one. That work asks how to *carry* uncertainty forward through the chain. This proposal asks a different post-training question: whether a frozen representation lets one attribute an observed failure to a declared family, abstain when it does not, and determine whether the resulting alarm corresponds to scientific consequence. The two are complementary — propagated uncertainty is a natural candidate covariate for the consequence axis here.

Deliberately small first experiment: one frozen checkpoint, one reproducible embedding, one or two physically valid perturbation families, clean plus three frozen severities, no retraining, one representation metric and one downstream physics metric.

Table 3: Candidate endpoints for a hierarchical-reconstruction arm.

Question	Possible endpoint
Embedding separability	Intra-/inter-cluster distance ratio, neighbour purity, or margin
Clustering robustness	ARI, efficiency, purity, fragmentation/merging
Perturbation stability	Neighbour retention; representation displacement
Near-vertex behaviour	Metrics binned by distance to the interaction vertex and local multiplicity
Calibration	Edge/cluster confidence vs observed clustering failure
Consequence	Particle- or interaction-level reconstruction metric

6 Collaboration

The NuBench and waveform arms provide independent groundwork. A collaboration could add two capabilities that the core study cannot supply: validation within a hierarchical reconstruction chain and a physically defensible detector-geometry contrast.

Table 4: Possible branches.

Branch	Initial joint decision	Minimal pilot
Representation diagnostics	Which perturbation and reconstruction stage are scientifically meaningful?	One Graph-SPICE embedding, one downstream clustering metric
Geometry robustness	Which geometry contrast and matching variables are defensible?	One paired simulation contrast
Core framing	Are the N/S boundary and the consequence formulation physically meaningful?	Protocol feedback only

Discussion requested. I would value Junjie's view on which of two extensions is scientifically stronger: (A) perturbation and near-vertex stability of SPINE/Graph-SPICE representations, or (B) paired detector-geometry robustness under a common acquisition intervention. If one direction appears worthwhile, the first step would be to jointly define one perturbation, one representation, and one downstream physics metric. Pipeline access or data exchange would only be discussed after that scientific scope is agreed.

7 Feasibility gate

Dataset access, checkpoint loading, released-prediction rescoring, intermediate-hook extraction, and a monotone module-dropout positive control have been validated ($15.28° \to 25.12°$ median angular error and 10-NN retention $1.000 \to 0.393$ across 0–50% dropout). Exact re-inference parity remains unresolved: the current pipeline differs from the released directions by 0.944° median angular error. All confirmatory re-inference experiments remain gated while event selection, preprocessing, graph construction, model restoration, and metric parity are checked stagewise on a fixed development subset. The tolerance will follow from the reproducibility requirement that diagnosis identifies, rather than being fixed in advance.

References

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